

# DEVA ANAND

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## Education

**MS, Computer Science**, University of Massachusetts Amherst 2025–2027  
*Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering*  
**BE, Computer Science and Engineering**, Vels Institute of Science, Technology and Advanced Studies 2019–2023

## Professional Experience

**TherAlign Health**, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer on a provider-facing **SMART on FHIR** clinical decision support app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that integrates multiple clinical data sources, **RxNorm** drug normalization, formulary/prior-authorization data, and PubMed/guideline evidence, into structured medication-alternative results, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

**Culture & Morality Lab, UMass Amherst**, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR (Contextualized Construct Representations) text-analysis platform (Python, **FastAPI**, sentence-transformers, React), turning a published NLP method into a self-serve pipeline that ingests research corpora and produces analysis-ready construct scores.
- Designed versioned research-data infrastructure, a YAML **model registry** as single source of truth, an append-only construct library (**99 constructs from 38 questionnaires**) with item hashes and immutable per-run snapshots, and a structured **data-quality** warning system, and grew the backend suite to **40 hermetic tests** running in CI without ML dependencies.

**Wysa**, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** by integrating data across services and Mayden iaptus **REST APIs**; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, replacing spreadsheet-based labeling with a structured workflow that cut manual annotation by **100+ hours/month** and lifted labeling throughput by **60%**.

## Projects

**Spark ETL Performance Optimization** (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a **PySpark** ETL + analytics pipeline over a ~1.4B-row NYC TLC dataset with broadcast joins, Adaptive Query Execution, skew-join handling, shuffle-partition tuning, schema-aware column-pruned reads, and year/month partitioning, cutting end-to-end ETL runtime ~**25%** (17–19 min to 14.3 min) and the tail-latency ratio **2.14x to 1.74x**; profiled stage timings in the Spark UI.

**AgenticSearch – Provenance-First Entity Discovery** (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage Python/**FastAPI** data pipeline that turns open-ended queries into **structured entity tables** via typed retrieval planning, page rerank, and deterministic extraction with LLM fallback, returning **cell-level provenance** (value, source, snippet, confidence) for every value; hardened with **150+ automated tests**.

**KG2RAG-Enhanced – Multi-Hop QA on HotpotQA** (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided **RAG** data pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR) on HotpotQA, lifting supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG<sup>2</sup>RAG baseline on a 4,905-question evaluation.

## Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

## Skills

**Languages & Databases:** Python, SQL, TypeScript, JavaScript, Java, C++, Bash, PostgreSQL, MongoDB, SQLite  
**Data & Pipelines:** PySpark, Spark, ETL, distributed data processing, data modeling, partitioning, data-quality checks, Hadoop / MapReduce  
**Backend & APIs:** FastAPI, Node.js, Express, Next.js, REST APIs, async services, pytest / integration testing  
**ML & AI:** PyTorch, sentence-transformers, Hugging Face Transformers, RAG, LLM agents, evaluation  
**Cloud & DevOps:** Azure, AWS, GCP, Docker, Git, GitHub Actions, CI/CD, Linux